Venkata Lakshmi Parimala Pasupuleti

Sr. AI Data Engineer · RAG Pipelines · ETL at Scale · Retrieval Evaluation · Python · AWS (S3 · Lambda · Bedrock · OpenSearch) · LangGraph · Vector Databases · Hybrid Search · Document Chunking · PII Compliance

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PROFESSIONAL SUMMARY

AI Data Engineer who builds RAG pipelines and the evaluation infrastructure to measure their quality — then iterates until the numbers improve. Built **AgentProbe**: open-source retrieval eval toolkit with 5-metric engine (precision, recall, groundedness, latency, hallucination rate), pass@k harness, golden datasets, and CI/CD regression gate that blocks pipeline changes when quality drops. Built **Leasel**: production RAG pipeline with section-aware **document chunking**, TF-IDF + semantic **hybrid search + reranking**, ChromaDB vector retrieval, and citation grounding to exact source clauses — evaluated across 1,240 documents (93.6% precision, 87.9% F1). 8 years of **ETL pipeline** ownership at Entrata: AWS S3/Glue ingestion, RabbitMQ async processing with idempotency guarantees, data quality validation, and **100% PHI compliance** for 100K+ housing units at 99.5% uptime. Currently building ML pipelines at Blue Horse Digital. M.S. Data Analytics Engineering, GMU (GPA 3.87).

WORK EXPERIENCE

Blue Horse Digital
Sep 2026 – Present

AI/ML Engineer (Volunteer) · Virginia, USA

- Building ML models and data pipelines; integrating AI components with back-end and front-end teams; surfacing business data insights into application APIs and reporting layers.

Magizhs Tech Solution
May 2025 – Aug 2025

AI/ML Engineer Intern — RAG · LangGraph · Document Ingestion · Compliance AI

- Built ingestion pipeline and RAG retrieval system (FAA Part 107 regulatory corpus) with document chunking, vector embedding, hybrid search, and citation grounding — every retrieved fact cited to exact source clause. Validation guardrails enforced compliance rules deterministically. Full ownership from ideation to deployed production system.

Entrata (PropTech / Real Estate SaaS)
Sep 2018 – Aug 2024

Senior Software Engineer — ETL Pipelines · AWS · Data Quality · RabbitMQ · PHI Compliance

- **Owned ETL ingestion pipeline for 7 years** — few thousand federal source rows transformed and loaded into 180,000 domain records powering eligibility, rent (HAP), and tax-credit calculations for 100K+ housing units at 99.5% uptime. Every release included data quality validation, reconciliation, and rollback preparation. Zero data quality failures or compliance violations across the entire ownership period.
- **Built RabbitMQ-based async ingestion system** for structured data updates at scale — retry logic and idempotency guarantees eliminating double-processing; owned through production incidents and annual cycles. Optimized 200+ SQL queries (30% gain) via execution plan analysis, indexing, and partitioning.
- **Led AWS S3 document migration with 100% PHI compliance** — Libsodium-encrypted pipelines, zero security incidents; implemented **PII handling and secure data isolation** across regulated housing data. Mentored 5+ engineers; 300+ code reviews; shipped 20+ production modules across 120+ Agile sprints.

George Mason University
Aug 2024 – Aug 2025

Research Associate & Teaching Assistant

- Built multi-source ETL pipelines (Python, R, 190+ countries); Isolation Forest anomaly detection across 3 data pipelines; $R^2 = 0.977$. Fine-tuned classification model (HuggingFace + PyTorch), deployed inference API. TA for 600+ students: Python, SQL, data modeling.

Entrata India, Pune
Jan 2017 – Sep 2018

Associate / Trainee Software Engineer

- ETL pipelines and compliance modules (Python, Java, PostgreSQL); led AWS S3 migration resolving 90% of legacy storage constraints; redesigned document storage architecture for hundreds of thousands of records.

PROJECTS

AgentProbe: Retrieval Evaluation Framework — Precision, Recall, Regression Gate, Golden Datasets
RETRIEVAL EVAL · PRODUCTION AI

Python · FastAPI (15 endpoints) · LangGraph · OpenAI · Anthropic · Groq · Ollama · ChromaDB · CI/CD (GitHub Actions) — github.com/venkatalakshmiparimala/AgentProbe

- Built fully local, zero-cost retrieval evaluation toolkit: **5-metric eval engine** (correctness, groundedness, latency, token efficiency, hallucination rate), **pass@k harness** with multi-trial sampling, CI-integrated **regression gate** with golden datasets (blocks pipeline changes when retrieval quality drops), model routing across OpenAI/Anthropic/Groq/Ollama. Offline + online evaluation loops. TDD-based: every pipeline change verified against benchmarks before merge.

Leasel: RAG Pipeline — Document Chunking, Hybrid Search + Reranking, Vector Retrieval, Citation Grounding
RAG PIPELINE · LLM EVAL

Python · LangGraph · FastAPI · ChromaDB · TF-IDF · OCR · SQLite — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Production RAG pipeline: OCR-fallback ingestion, **section-aware document chunking**, **hybrid search (TF-IDF + semantic reranking)**, ChromaDB vector retrieval, citation grounding to exact source clauses, missing/conflicting clause detection. Knowledge store built over 1,240 residential lease PDFs (5 US states + federal). Evaluated via AgentProbe: **93.6% precision, 87.9% F1, pass@3 88.3%**. Continuously iterated chunk size and overlap parameters to improve retrieval quality.

AI on FHIR: EHR Data Ingestion + NLP Retrieval Platform — HIPAA, PII Compliance
DATA ENGINEERING · HIPAA · PII

Flask · LLMs · FHIR API · BERT · Embeddings · spaCy · React · Docker — github.com/venkatalakshmiparimala/AI_on_FHIR

- Ingestion and retrieval platform over FHIR EHR APIs — NLP queries (EN/ES) translated to structured clinical data retrieval using LLMs, BERT, and embeddings; **HIPAA-compliant RBAC with PII data isolation**. ~60% reduction in manual retrieval time. Production-deployed, Docker containerized.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia
May 2026 · GPA 3.87

Data engineering, ML systems, statistical modeling, AI pipelines, data governance, anomaly detection

B.Tech Computer Science & Engineering — JNTU Anantapur
2013 – 2017

TECHNICAL SKILLS

RAG & Retrieval: RAG Pipelines, Document Chunking, Semantic Chunking, Vector Embeddings, Hybrid Search, Reranking, Contextual Retrieval, Ingestion Pipelines, Knowledge Stores, ChromaDB, Qdrant, Amazon OpenSearch (adjacent), Citation Grounding, LLM Evaluation, LLM-as-Judge, Retrieval Precision/Recall/Relevance, Golden Datasets, Regression Harnesses, Pass@k Eval, TDD, Test-Driven Development, Offline Evaluation, Online Evaluation, A/B Testing, Hallucination Detection, Confidence Scoring, Guardrails

AWS & Cloud: AWS S3, EC2, RDS, Glue, Lambda, Step Functions, AWS Bedrock, Amazon OpenSearch; Azure (ADF, ADLS, Databricks); GCP; Docker, Kubernetes, OpenShift, CI/CD (GitHub Actions), Terraform, Prometheus, Grafana, OpenTelemetry

Data Engineering & ETL: ETL/ELT Pipelines, Data Ingestion at Scale, Data Quality Frameworks, Validation, Data Completeness, RabbitMQ, Idempotency, Async Processing, Apache Spark, PySpark, Data Modeling, Data Lineage, Data Governance, Performance Tuning, PostgreSQL, MySQL, Oracle, Snowflake, Teradata, SQL Server (T-SQL), Solr, NoSQL

AI & Agents: LangChain, LangGraph, Agentic AI, Multi-Agent Orchestration, Tool Calling, Function Calling, Structured Outputs, HITL, Agent Observability, Prompt Engineering, OpenAI API, Anthropic API, Groq, Ollama, Production AI, LLMOps, MCP

Security & Compliance: PHI Compliance, PII Handling, HIPAA RBAC, Data Isolation, Libsodium Encryption, Regulatory Compliance, Secure Coding Practices, Audit Trails

Languages & Backend: Python, Java, JavaScript, TypeScript, PHP, Scala, SQL, R, Bash; FastAPI, Spring Boot, REST APIs, Microservices, Distributed Systems

ML: PyTorch, TensorFlow, Scikit-learn, HuggingFace, XGBoost, Isolation Forest, Pandas, NumPy, spaCy, BERT, NLP, Anomaly Detection